

Jeremy Guido

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Professional Summary

Mechanical engineer with 9+ years of professional experience across energy consulting, applied research, and industrial refrigeration design. Skilled in building energy modelling, mechanical system design, instrumentation, and technical documentation. Hands-on background in fabrication, embedded systems, and controls from applied research. Comfortable working across field, office, and lab environments.

Featured Project

Distributed Backflow Preventer Monitoring System

Camosun Innovates

- Designed and deployed a remote monitoring system to detect leaks and failure modes in live backflow preventer installations.
- Selected sensors; implemented fault detection logic, data acquisition, and gateway communication to a web-based platform.
- Performed field deployment, commissioning, functional testing, power optimisation, and technical documentation.
- Deployed 4 units at a single facility; maintained continuous data collection with minimal downtime.

Work Experience

Polar Engineering, Victoria, BC

Aug 2023 – Jan 2026

Project Engineer

- Conducted energy audits and feasibility studies; developed retrofit concepts for refrigeration and low-carbon systems.
- Built energy and emissions models; produced technical reports and grant documentation for client deliverables.
- Reviewed mechanical and electrical drawings; coordinated vendor and contractor inputs across projects.
- Assessed industrial refrigeration and heat pump systems; supported projects from site investigation through final report.

Focal Engineering, Victoria, BC

Jun 2022 – Jul 2023

Mechanical Engineer

- Developed and calibrated building energy models; supported mechanical system design and calculations.
- Produced construction drawings and specifications; maintained energy databases and technical summaries.
- Performed mechanical load calculations and supported equipment selection for HVAC and ventilation systems.
- Coordinated with project teams to align energy model assumptions with design drawings and code requirements.

Camosun Innovates, Victoria, BC

Jan 2016 – Jun 2022

Applied Research Technologist II

- Technical lead on applied research in mechanical systems, automation, sensing, and embedded controls.
- Designed and tested instrumented prototypes using CAD/CAM, CNC machining, laser cutting, and rapid prototyping.
- Developed ESP32 firmware and LoRa mesh networking for battery-powered remote monitoring in field environments.
- Produced CAD drawings, schematics, and documentation; supported deployment, commissioning, and iterative refinement.

Accent Refrigeration, BC

Jun 2016 – Jul 2018

Design Technologist

- Designed industrial refrigeration packages and coordinated fabrication and installation.
- Developed mechanical layouts for ice rink and ice plant systems; performed equipment selection and load calculations.
- Produced P&ID, mechanical, and electrical drawings in accordance with industry standards.
- Conducted site visits and supported system upgrades to meet regulatory and operational requirements.

Education

University of Victoria, Victoria, BC

Sept 2018 – Dec 2021

Bachelor of Engineering

Camosun College, Victoria, BC

Sept 2015 – May 2017

Mechanical Engineering Technologist

Okanagan College, Kelowna, BC

Jan 2008 – Jul 2012

Red Seal Chef

Relevant Skills

Programming

Python, C, C++, MATLAB

Controls and Embedded

ESP32, sensor integration, control logic, data logging

CAD and Documentation

AutoCAD, SolidWorks, Inventor, P&IDs, technical reporting

Manufacturing and Prototyping

CNC machining, laser cutting, waterjet cutting, rapid prototyping

Tools and Platforms

MS Office, L^AT_EX

Credentials

Valid driver's license